

SULIT



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER 2, 2024/2025

SUBJECT CODE : SECP 3213
SUBJECT NAME : BUSINESS INTELLIGENCE
YEAR/COURSE : 3/SECPH
TIME : 3 HOURS
DATE/DAY :
VENUE :

INSTRUCTIONS TO THE STUDENTS:

1. For Section A, write the answer on answer booklet.
2. For Section B and C, saved the file in the allocated folder at your PC.
3. A candidate who is suspected of cheating in examination is liable to disciplinary action including (but not limited to) suspension or expulsion from the university. All materials and/or devices which are found in violation of any examination rules and regulation will be confiscated.

NAME	
MATRIC NO.	
SECTION	
LECTURER	

(This question paper consists of 7 pages including this page)

SECTION A

[10 MARKS]

Select the correct answer.

1. What is the main goal of Business Intelligence?
 - A. To collect customer feedback
 - B. To display static charts for printing
 - C. To turn raw data into meaningful insights for decision-making
 - D. To store files on a shared drive
2. In the context of Business Intelligence, what does data granularity refer to?
 - A. The level of detail or depth of data stored and analyzed
 - B. The physical location where the data is hosted
 - C. The amount of disk space required for storing data
 - D. The time duration for which data is archived
3. Which type of Business Intelligence analytics is used to recommend optimal decisions or actions?
 - A. Descriptive analytics.
 - B. Prescriptive analytics.
 - C. Predictive analytics.
 - D. Diagnostic analytics
4. What is the purpose of the Input Data tool in Alteryx?
 - A. To export data into an Excel file
 - B. To apply calculations to numeric fields
 - C. To bring external data files into the workflow
 - D. To create visual dashboards
5. Which of the following tasks can be performed using the Formula tool?
 - A. Remove duplicate rows
 - B. Join two tables
 - C. Create a new calculated field like $\text{Revenue} = \text{Quantity} \times \text{UnitPrice}$
 - D. Generate a report in PDF format
6. Which statement best describes the Union Tool?
 - A. It compares numeric relationships between fields
 - B. It stacks multiple datasets vertically (adds rows)
 - C. It removes duplicate values across datasets
 - D. It joins datasets based on shared field values

7. What is the main purpose of the Field Summary tool in Alteryx?
 - A. To visualize relationships between multiple tables
 - B. To identify missing values, data types, and summary statistics
 - C. To generate pivot tables for grouped values
 - D. To apply clustering to fields
8. Which type of result does the Association Analysis tool provide?
 - A. A timeline of changes in a field
 - B. A regression coefficient list
 - C. Rules showing relationships between field values
 - D. A time series forecast
9. Which of the following best describes the main purpose of Tableau?
 - A. Data cleaning and ETL processing
 - B. Data visualization and dashboard creation
 - C. Predictive modeling and statistical analysis
 - D. File conversion and automation scripting
10. From your understanding, how can Tableau and Alteryx work together in a business workflow?
 - A. Tableau can clean data and export to Alteryx
 - B. Alteryx builds dashboards and Tableau runs SQL queries
 - C. Alteryx prepares and processes data, which Tableau then visualizes
 - D. Tableau and Alteryx are used only in cloud environments

SECTION B

[50 MARKS]

For this section, you need to analyze data from the airline industry by design a workflow using Alteryx Designer. Your final output is a Flight Status Report as shown in **Figure 5**. You need to conduct the following task (step by step) as a guideline to complete your solution.

1. Read all datasets below using one Input Data tool.
 - *Flight Data 1. yxdb*
 - *Flight Data 2. yxdb*
 - *Flight Data 3. yxdb*(6 marks)
2. Read the data – *Airline Code Lookup.csv*. This file contains 14 airline names and their associated two-letter codes. Connect the flight data above and the airline code lookup file to the Find Replace tool to change the airline code into the full name for flight data.(4 marks)
3. Connect a formula tool to add 2 news columns: LONG DELAY and SHORT DELAY and configure it as shown in Figure 1.

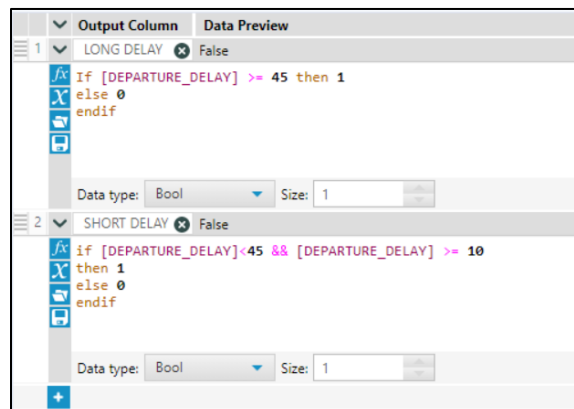


Figure 1

- (5 marks)
4. You must calculate flight delay rates for each airline. To calculate these percentages, you need to aggregate the values at the airline level for the number of short delays, long delays, and total flights. To get these three aggregations, you need three separate summarize tools:
 - summarize tool to count total AIRLINE.
 - summarize tool to count LONG DELAY by AIRLINE.
 - summarize tool to count SHORT DELAY by AIRLINE.

(9 marks)

5. Looking at the results of the long and short delay aggregations, you will see that you have the totals for flights with and without delays. Use Filter tool to filter only delay flight (Long or Short delay). Figure 2 show example of configuration.

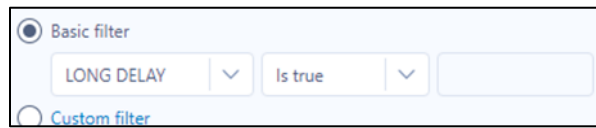


Figure 2

(4 marks)

6. Join the result from summarize tool (step 4) with the result in step 5 to combine both information in one table. Figure 3 shows an example of output.

Results - Join (12) - Out - Join

4 of 4 Fields ☒ 6 records displayed ✖ ✓

Record	AIRLINE	LONG DELAY	LONG DELAY COUNT	All Flights
1	United	True	1242	12199
2	Delta	True	2577	20612
3	American	True	968	12932
4	US Airways	True	1145	9807
5	Southwest	True	2361	29524
6	Virgin America	True	102	1377

Figure 3

You should add separate join tool for short delays.

(8 marks)

7. Add Formula tool to calculate the percentage of flight delay. Figure 4 shows an example of expression.

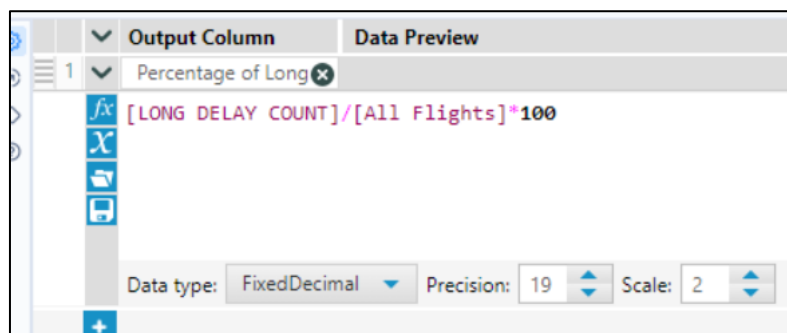


Figure 4

(4 marks)

8. Join the table of Long Delay and Short Delay and create a final report as shown in Figure 5. The report date will be the latest based on your PC's settings.

Flights Delay Status			alteryx
2025-05-17 13:41:18			
AIRLINE	Percentage of Long Delays	Percentage of Short Delays	
American	7.49	15.65	
Delta	12.50	21.13	
Southwest	8.00	21.70	
United	10.18	25.50	
US Airways	11.68	18.08	
Virgin America	7.41	14.09	

Figure 5

(10 marks)

SECTION C**(40 MARKS)**

For this section, you need to create a workbook in Tableau Desktop. Load the dataset *GooglePlayStore.xlsx* into your new workbook. **Solution for each question must be visualized at different worksheet and rename it based on question no.** (Example: Q1)

1. Create a **dual-axis chart** using **Review** and **Rating** measures by **Category**. Use bar graph to denote the review whereas line graph to denote the rating. Sort by Review in descending order. Determine the categories that have high review but low rating and state the value for the review and rating by completing Table 1.

Table 1

Category	Review	Rating

(5 marks)

2. Show the top 3 app based on reviews for **Android version 0.0.9**. Use a context filter for the Current Version field as a second filter by right clicking on the Current Version filter and select Add to context. **(6 marks)**
3. Create a **treemap chart** to visualize the number of installation based in genres. Add category information in the tooltip. Identify the genre and its category with the highest installation for paid status **(4 marks)**
4. Create a **set** of apps with rating **higher than 4.5**. Determine the **percentage** of apps that fall into this set. **(6 marks)**
5. Use the count measure to determine the number of records in the Social category for both free and paid status. **(4 marks)**
6. Which content rating category has the highest review? **(3 marks)**
7. Create a chart to sort categories based on percentage of installs. Determine the percentage of install for lifestyle category with rating of at least 5. **(6 marks)**
8. Change the data type for last update filed to date data type and create a chart to evaluate the monthly trend of number of app installation (used count measurement) from May 2010 until July 2018. Forecast the number of app installs for May 2019.

(6 marks)***End of Questions***